SECTION VIII: CURRICULUM VITAE (SINGAPORE LEAD PI)

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5. Current position and past employment history

Period (MM YYYY – MM YYYY)	Organisation, Country	Position	Appointment type	Research topic	Research supervisor (if applicable)
04 2024 – Present	Structural Materials Department, Institute of Materials Research and Engineering (IMRE), A*STAR, Singapore	Senior Scientist	Full-time **[CHECK: Permanent / Contract]**	Group leadership and Lead PI/Co-PI in active infrared thermography (induction, optical flash/pulsed, lock-in, vibro-, and laser IRT) and automated NDE of aerospace CFRP/GFRP composite structures. Co-PI on Singapore Aerospace Programme (SAP) developing induction thermography and 3D multiphysics FEM electromagnetic-thermal simulations for composite repair. Lead researcher on physics-informed deep learning for automated defect recognition (ADR) on curved aeronautical composites, and multimodal NDE cross-validated against ultrasonic testing (UT), X-ray CT, and shearography. Project member on AI Singapore's MIDGAI program. Technical Committee Member of ISO/TC 135/SC 8 (Thermographic Testing) and Member of ASTM International.	NA (Independent PI; Dept. Head: Dr. Andrew Ngo Chun Yong **[CHECK: confirm if NA preferred]**)
01 2026 **[CHECK: confirm start month]** – Present	Future Energy Acceleration and Translation (FEAT) Centre, IMRE, A*STAR, Singapore	Senior Scientist	Joint Appointment	Lead NDE researcher on AMOFES and STRIDE projects: optical pulsed laser thermography, thermal modeling, and quantitative interfacial thermal resistance characterization of advanced coatings and extreme-environment structural materials.	NA
11 2025 – 02 2026	Mechanical Design and Machine Construction (IIND-03/A) NDT Research Group, Politecnico di Bari, Italy	Visiting Researcher	Visiting Fellowship	Bilateral Italy–Singapore research collaboration on multi-layer thermal signature analysis and machine learning algorithms for automated in-line thermographic defect detection, sizing, and characterization.	Prof. Umberto Galietti / Dr. Davide Palumbo **[CHECK: confirm host supervisor]**
11 2021 – 03 2024	Structural Materials Department, Institute of Materials Research and Engineering (IMRE), A*STAR, Singapore	Scientist	Full-time	Work Package Lead on Polymer Matrix Composites Programme (PMCP) for in-line IRT monitoring of composite manufacturing, induction and optical (pulsed/lock-in) thermography of impact and manufacturing defects in aerospace CFRP laminates, honeycomb sandwich structures, and filament-wound GFRP pipes. Principal Investigator (PI) on multinational collaborative grant developing quantitative active thermography, FEM simulation, and 3D optical profile correction for complex curved aerospace composite structures.	NA (PI / WP Lead; Group Leader: Dr. Andrew Ngo Chun Yong **[CHECK: confirm]**)

Period (MM YYYY – MM YYYY)	Organisation, Country	Position	Appointment type	Research topic	Research supervisor (if applicable)
05 2018 – 09 2021	Centre for Non-Destructive Evaluation (CNDE), Indian Institute of Technology Madras (IIT Madras), India	Senior Project Officer	Full-time	Induction heating thermography of CFRP composite laminates; optical pulsed and laser-assisted IRT for coatings and surface-breaking crack inspection; convolutional neural network (CNN) deep learning for automated defect recognition (ADR) on X-ray radiographs and ray-tracing X-ray/CT simulation; multiphysics FEM modeling of anisotropic heat diffusion in composite microstructures. Co-supervised 4 Ph.D. and 2 M.Tech. researchers.	Prof. Krishnan Balasubramaniam & Prof. C. V. Krishnamurthy
02 2020 – 05 2020	Entrepreneur First (EF), Bangalore, India	Cohort Member (BA3)	Entrepreneurial Fellowship **[CHECK: confirm type]**	Deep-tech translation and prototype validation of cloud-based spectral/imaging hardware and AI algorithms for automated non-destructive quality inspection.	NA
01 2016 – 06 2017	NDT Group 8.7 (Thermographic Methods), Federal Institute for Materials Research and Testing (BAM), Berlin, Germany	Postdoctoral Research Scientist	Full-time (Postdoctoral)	Project VITCEA (Validated Inspection Techniques for Composites in Energy Applications): quantitative optical active thermography (pulsed/flash, lock-in, and step-heating IRT) and computational FEM heat-transfer modeling for defect characterization in fiber-reinforced polymer (FRP) composite structures for wind energy and aerospace/industrial applications; laser thermography for crack detection.	Dr. Christiane Maierhofer / Dr. Mathias Ziegler **[CHECK: confirm BAM supervisor]**
01 2015 – 03 2015 **[CHECK: 03 2015 or 06 2015]**	Electromagnetic NDT Research Group, West Pomeranian University of Technology (ZUT), Szczecin, Poland	Visiting Research Scientist	Visiting Fellowship (EU FP7 Secondment)	EU FP7 Project HEMOW (Health Monitoring of Offshore Wind Farms): active infrared thermography, FEM numerical modeling of heat-sink effects, and THz time-domain NDE for defect detection in composite wind turbine blades.	Prof. Tomasz Chady / Dr. Barbara Szymanik **[CHECK: confirm ZUT supervisor]**
09 2014 – 12 2015	Centre for Non-Destructive Evaluation (CNDE), Indian Institute of Technology Madras (IIT Madras), India	Institute Postdoctoral Fellow	Full-time (Postdoctoral)	Multiphysics FEM modeling of microstructural heat diffusion and elastic wave propagation in heterogeneous materials; in-line laser and active IR thermography for automated process monitoring and crack detection; co-supervision of Ph.D. and Master students.	Prof. Krishnan Balasubramaniam & Prof. C. V. Krishnamurthy
04 2007 – 12 2008	Centre for Non-Destructive Evaluation (CNDE), Indian Institute of Technology Madras (IIT Madras), India	Research Associate	Full-time	Automated online infrared thermography, X-ray radiography validation, and finite element thermal modeling for aerospace AA2219 structural weld inspection.	Prof. Krishnan Balasubramaniam & Prof. C. V. Krishnamurthy

6. Academic qualifications

- **Ph.D. in Mechanics & Civil Engineering**, Department of Mechanics and Civil Engineering (LMGC), University of Montpellier 2 (UM2), Montpellier, France — **Year Awarded:** 2014
  Topic: Weld pool shape identification using Multi-physics modelling and Experiments (*Static GTAW: experimental and numerical investigations and heat flux parameter estimation*).
- **M.S. (by Research) in Mechanical Engineering**, Department of Mechanical Engineering, Indian Institute of Technology Madras (IIT Madras), Chennai, India — **Year Awarded:** 2011 **[CHECK: 2010 or 2011 on degree scroll]**
  Topic: Online weld quality monitoring using Infrared Thermal Imaging.
- **Bachelors in Mechanical Engineering (AMIE)**, Department of Mechanical Engineering, Institution of Engineers (India), India — **Year Awarded:** 2006

Topic: Mechanical Engineering.

7. Research interests

Physics-informed artificial intelligence and deep learning for automated defect detection, sizing, and confidence estimation in non-destructive testing (NDT); active infrared thermography combining electromagnetic induction and optical (flash/pulsed, lock-in, and laser) excitation for polymer matrix composites (CFRP/GFRP), curved aerospace structures, and wind turbine blades; multiphysics finite element modeling (FEM) of electromagnetic-thermal and optical heat transfer in anisotropic composites to generate synthetic training datasets; automated, in-line, and robot-assisted thermographic inspection systems with multi-sensor data fusion and 3D defect mapping; and multimodal NDE validation using ultrasonic testing (UT), X-ray computed tomography (CT), shearography, and terahertz/hyperspectral imaging.

8. Publications in the last 3 years (Nov 2023 – Nov 2026)

Peer-Reviewed Journal Articles (Nov 2023 – Nov 2026)

J1. **Sreedhar Unnikrishnakurup**, S. Sampath, Z. W. Tham, S. W. Tay, J. Zheng, V. Kumar, W. Thitsartarn, L. Zhang, and A. Ngo. "Multimodal non-destructive evaluation of lightning strike damage in painted CFRP laminates: An infrared thermography-based damage index approach with cross-validation." *NDT & E International*, Vol. 161, 103817, 2026. URL/DOI: https://www.sciencedirect.com/science/article/abs/pii/S096386952600188X **[CHECK: insert DOI string]**. **[Impact Factor: 4.1 (2023 JCR) [CHECK: verify 2024 JCR IF]]**

J2. **Sreedhar Unnikrishnakurup**, R. K. T. Kidangan, C. V. Krishnamurthy, K. Balasubramaniam, and A. Ngo. "Assessment of fiber orientation and order in carbon fiber reinforced polymer composites using induction thermography and Radon transform analysis." *Composites Part B: Engineering*, Vol. 296, 112224, 2025. URL: https://bit.ly/4c1dzsB **[CHECK: insert full DOI]**. **[Impact Factor: 12.7 (2023 JCR) / 14.2 (2024 JCR) [CHECK: confirm JCR year]]**

J3. L. Bounenni, M. Arbane, C. Ibarra-Castanedo, Y. Yaddaden, **Sreedhar Unnikrishnakurup**, A. N. C. Yong, and X. Maldague. "Advanced defect detection on curved aeronautical surfaces through infrared imaging and deep learning." *NDT*, Vol. 2(4), pp. 519–531, 2024. URL: https://bit.ly/4hi8sW8 (DOI: https://doi.org/10.3390/ndt2040031). **[Impact Factor: NA / Emerging Sources Citation Index [CHECK: confirm if 2024/2025 JCR IF assigned]]**

J4. R. K. T. Kidangan, **Sreedhar Unnikrishnakurup**, C. V. Krishnamurthy, and K. Balasubramaniam. "Uncovering the hidden structure: A study on the feasibility of induction thermography for fiber orientation analysis in CFRP composites using 2D-FFT." *Composites Part B: Engineering*, Vol. 269, 111107, 2024 **[CHECK: confirm publication date $\geq$ Nov 2023]**. URL: https://bit.ly/4igfbB6. **[Impact Factor: 12.7 (2023 JCR) [CHECK: confirm JCR year]]**

J5. R. K. T. Kidangan, **Sreedhar Unnikrishnakurup**, C. V. Krishnamurthy, and K. Balasubramaniam. "Induction thermography for unidirectional CFRP composites: A novel inspection approach through global current path integration." *Composite Structures*, Vol. 327, 117678, 2024 **[CHECK: confirm publication date $\geq$ Nov 2023]**. URL: https://bit.ly/41wBBXL. **[Impact Factor: 6.3 (2023 JCR) [CHECK: confirm JCR year]]**

J6. **Sreedhar Unnikrishnakurup**, Z. Zhang, D. H. L. Seng, Z.-Q. Zhang, J. Pan, V. Kumar, J. Zheng, C. Manzano, and A. Ngo. "Exploring thermal dynamics and porosity of cold-sprayed Ti-6Al-4V coatings on Al6061-T6 substrates: A pulsed thermography and numerical modeling approach." *International Journal of Thermal Sciences*, Vol. 196, 108732, 2024 **[CHECK: confirm publication date $\geq$ Nov 2023]**. URL: https://bit.ly/4kDFovc. **[Impact Factor: 4.9 (2023 JCR) [CHECK: confirm JCR year]]**

Conference Papers (Nov 2023 – Nov 2026)

C1. **Sreedhar Unnikrishnakurup**, K. Wee, S. Sampath, and A. Ngo. "Enhancing low-energy impact damage detection in carbon composites using sweep-excited vibrothermography and machine learning." *The 18th International Conference on Quantitative InfraRed Thermography (QIRT 2026)*, July 2026.

C2. E. D'Accardi, **Sreedhar Unnikrishnakurup**, D. Palumbo, A. Ngo, and U. Galietti. "Multi-layer thermal feature analysis for defect detection and characterization in AlSi10Mg parts produced by laser powder bed fusion." *The 18th International Conference on Quantitative InfraRed Thermography (QIRT 2026)*, July 2026.

C3. **Sreedhar Unnikrishnakurup**, V. Kumar, J. Zheng, N. Wong, S. J. Wang, and A. Ngo. "Pulsed laser thermography for interfacial thermal resistance characterisation of tungsten coatings on boron carbide

substrates." *The 18th International Conference on Quantitative InfraRed Thermography (QIRT 2026)*, July 2026.

C4. **Sreedhar Unnikrishnakurup**, M. Sharma, A. Feng, J. Zheng, V. Kumar, C. Manzano, Z. Wong, and A. Ngo. "Lock-in thermography and machine learning-based defect detection in carbon fiber reinforced thermoplastics." *Singapore International NDT Conference & Exhibition (SINCE 2025)*, *e-Journal of Nondestructive Testing*, February 2025. URL: https://www.ndt.net/?id=30843.

C5. **Sreedhar Unnikrishnakurup**, R. T. Kidangan, C. V. Krishnamurthy, and K. Balasubramaniam. "Characterizing fiber orientations and layer stacking sequences in CFRP laminates through induction heating patterns: a nondestructive approach." *Proceedings of the 4th Asian Quantitative InfraRed Thermography Conference (QIRT-Asia 2023)*, 2023 **[CHECK: confirm proceedings/conference month $\geq$ Nov 2023; remove if earlier]**. DOI: https://doi.org/10.21611/qirt.2023.17.

C6. S. O. A. Gnessougou, **S. Unnikrishnakurup**, A. Ngo Chun Yong, P. Servais, C. Ibarra-Castanedo, and X. P. V. Maldague. "Curved CFRP specimen: Thermographic inspection and simulation." *Proceedings of the 4th Asian Quantitative InfraRed Thermography Conference (QIRT-Asia 2023)*, 2023 **[CHECK: confirm proceedings/conference month $\geq$ Nov 2023; remove if earlier]**. DOI: https://doi.org/10.21611/qirt.2023.22.

9. Patents held (related or unrelated to the study)

- **Granted Patents:** NA **[CHECK: if any of the 12 A*STAR Technology Disclosures have been filed as patent applications, insert application/publication numbers here; otherwise keep as NA / Technology Disclosures below]**.
- **Technology Disclosures & Industrial Licensing:** Filed **12 Technology Disclosures (TDs)** at the Institute of Materials Research and Engineering (IMRE), A*STAR (2021–2026) in the areas of active infrared thermography, composite defect inspection, and laser-based NDT, with **1 TD licensed to an industry partner** for composite inspection workflows in repair applications.

10. Scientific awards

- **Conference Co-Chair**, Asian Quantitative Infrared Thermography Conference (QIRT-Asia 2027), Singapore (2027).
- **Invited Keynote Speaker**, 2027 Formosa Workshop on Non-Destructive Testing (FWNDT 2027), Society for Nondestructive Testing & Certification of Taiwan (SNTCT), Taipei, Taiwan (2027).
- **Invited Speaker Recognition**, NDE 2026 Conference & Exhibition, Indian Society for Non-Destructive Testing (ISNT), Hyderabad, India (2026).
- **Visiting Researcher Fellowship**, Mechanical Design and Machine Construction (IIND-03/A) NDT Research Group, Politecnico di Bari, Italy (2025–2026).
- **Technology Commercialization & Licensing Recognition** (1 Technology Disclosure Licensed to Industry, 12 Technology Disclosures Filed), IMRE, A*STAR, Singapore (2021–2026).
- **International Standards & Scientific Steering Committee Appointments**: Technical Committee Member of ISO/TC 135/SC 8 (Thermographic Testing); Member of ASTM International; Steering Committee Member of the International Quantitative InfraRed Thermography (QIRT) Committee and Asian Quantitative Infrared Thermography (QIRT-Asia) Committee (2023–Present); Level 2 Certified in Infrared Thermography (Oct 2026).
- **Competitive Research Grant Leadership**: Co-Principal Investigator (Co-PI), Singapore Aerospace Programme (SAP) Research Grant on Induction Thermography for Composite Repair, A*STAR (2022–2026); Principal Investigator (PI), Multinational Collaborative Grant on Quantitative Thermographic Inspection of Curved Aerospace Composite Structures, A*STAR (2022–2025); Work Package Lead, Polymer Matrix Composites Programme (PMCP), A*STAR (2021–2025).
- **Visiting Researcher Fellowship**, EU FP7 Project: Health Monitoring of Offshore Wind Farms (HEMOW), West Pomeranian University of Technology (ZUT), Poland (2015).
- **Institute Postdoctoral Research Fellowship**, Indian Institute of Technology Madras (2014–2015) and **French Doctoral Research Allowance**, University of Montpellier 2, France (2011–2013).

11. Percentage of time spent in Singapore every year

100%